

DIGITAL LOGIC DESIGN

(CE_118)

CHAPTER 2:

Storage Components *(part_2)*

Storage components

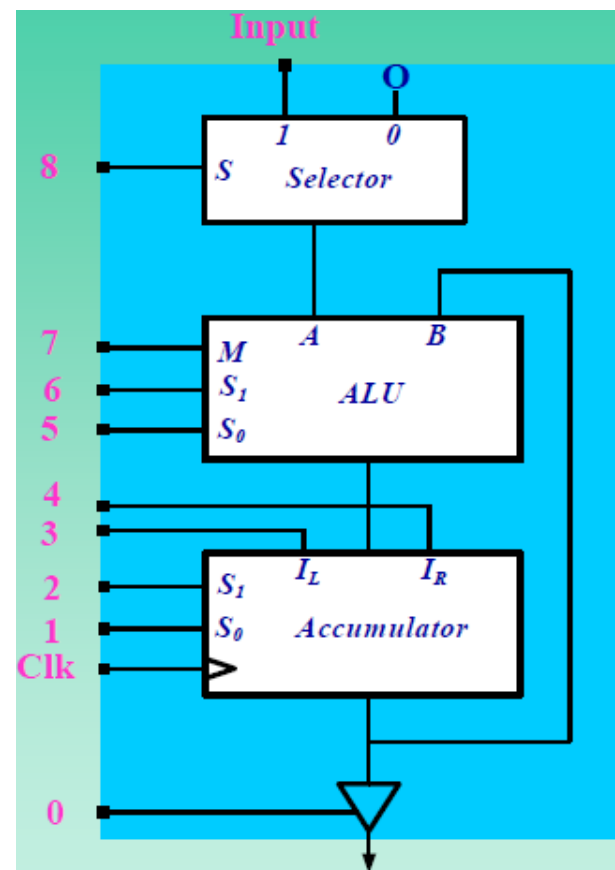
- Storage components store data and perform some simple operations.
- Storage components include:
 - registers
 - counters
 - memories
 - queues
 - stacks
- Combinatorial and storage components are used for construction of:
 - datapaths
 - controllers
- Main subsystems of modern processors and other microchips.

Simple datapath with one accumulator

- Datapath are used to perform complex numerical computation or data manipulation.
- Datapath consists of temporary storage in addition to arithmetic, logic, and shift unit.

```

sum = 0
loop:
  for i = 1 to 100
    sum = sum + xi
  end loop
  
```

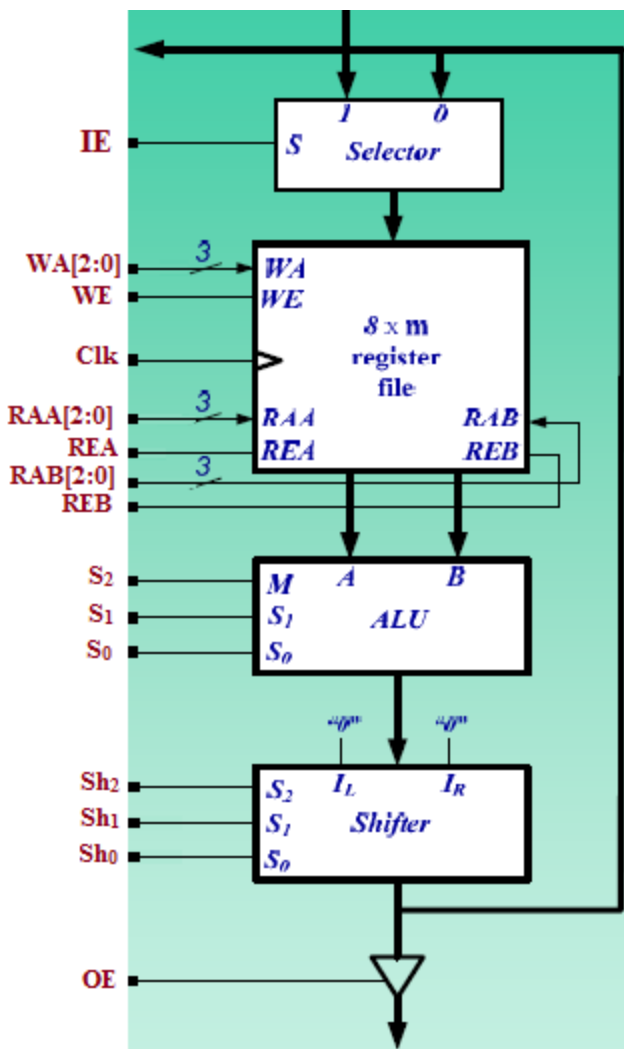


Datapath schematic

8	7	6	5	4	3	2	1	0
Input select	ALU controls			Shift values		Accumulator controls		Out enable

Control word

Datapath with 3 port register-file



Datapath
schematic

<i>M</i>	<i>S</i> ₁	<i>S</i> ₀	ALU Operations
0	0	0	Complement A
0	0	1	AND
0	1	0	EX-OR
0	1	1	OR
1	0	0	Decrement A
1	0	1	Add
1	1	0	Subtract
1	1	1	Increment A

ALU operations

<i>S</i> ₂	<i>S</i> ₁	<i>S</i> ₀	Shift Operations
0	0	0	Pass
0	0	1	Pass
0	1	0	Not used
0	1	1	Not used
1	0	0	Shift left
1	0	1	Rotate left
1	1	0	Shift right
1	1	1	Rotate right

Shifter operations

Control word

19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IE	Write address				Read address A				Read address B				ALU operation			Shifter operation			OE

One's-count algorithm

oExample: One's-counter implementation.

oProblem: Using a datapath with a 3 port register file, design a one's counter that will count the number of ones in an input dataword , and return the result after completion

One's-count algorithm

```

1. Data := Inport
2. Ocount := 0
3. Mask := 1
while Data ≠ 0 repeat
  4. Temp := Data AND Mask
  5. Ocount := Ocount + Temp
  6. Data := Data >> 1
end while
7. Outport := Ocount
Basic algorithm for one's counter

```



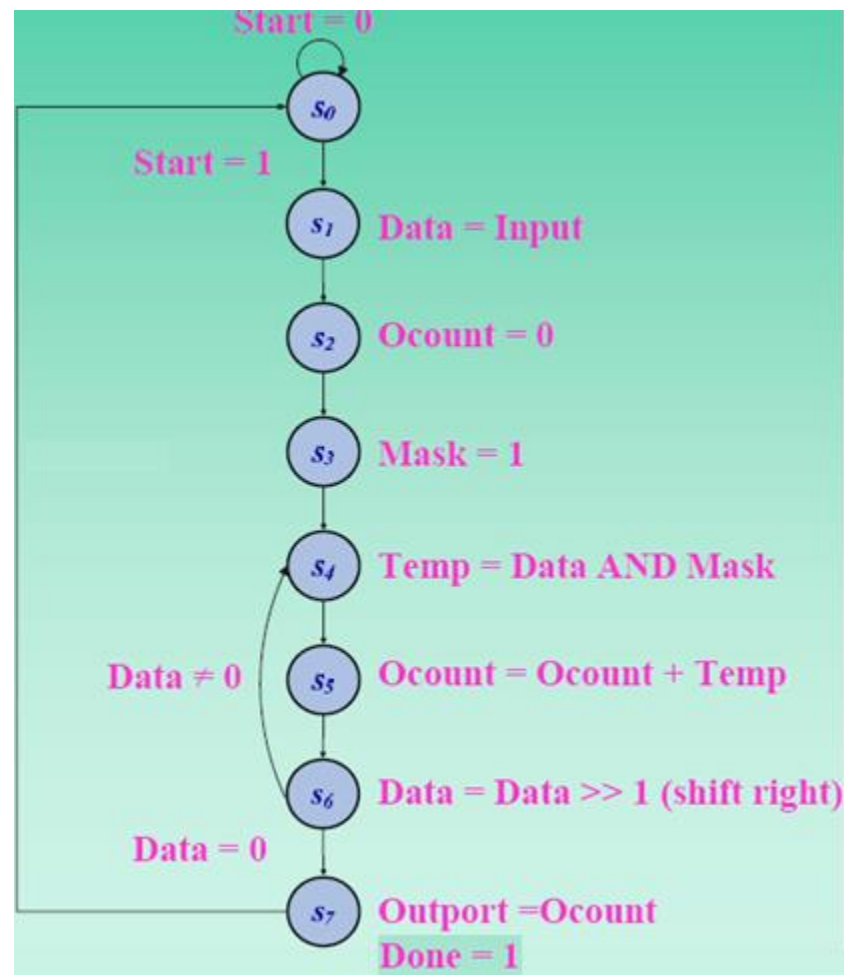
Control words	IE	Write address	Read address A	Read address B	ALU operation	Shifter operation	OE
1	1	R_1	X	X	X	X	0
2	0	R_3	0	0	Add	Pass	0
3	0	R_2	0	X	Increment	Pass	0
4	0	R_4	R_1	R_2	AND	Pass	0
5	0	R_3	R_3	R_4	Add	Pass	0
6	0	R_1	R_1	0	Add	Shift right	0
7	0	None	R_3	0	Add	Pass	1

Repeated
while
Data ≠ 0

Control words for one's counter

FSM representation of One's-counter

- State lasts for a clock cycle.
- In each state the datapath executes the statement indicated on its right side.



Next-state logic for One's- counter

States	$Q_2Q_1Q_0$	Start, (Data = 0)			
		00	01	10	11
s_0	0 0 0	000	000	001	001
s_1	0 0 1	010	010	010	010
s_2	0 1 0	011	011	011	011
s_3	0 1 1	100	100	100	100
s_4	1 0 0	101	101	101	101
s_5	1 0 1	110	110	110	110
s_6	1 1 0	100	111	100	111
s_7	1 1 1	000	000	000	000

Next-state table

$$Q_2(next) = Q_2'Q_1Q_0 + Q_2Q_1' + (Data=0)Q_2Q_1'Q_0'$$

$$Q_1(next) = Q_1'Q_0 + Q_2'Q_1Q_0' + (Data=0)Q_1Q_0'$$

$$Q_0(next) = Q_2'Q_1Q_0' + Q_2Q_1'Q_0' + StartQ_2'Q_0' + (Data=0)Q_2Q_0'$$

Next-state equations

Start, (Data = 0)		$Q_2 = 0$				$Q_2 = 1$			
		Q_1Q_0				Q_1Q_0			
		00	01	11	10	00	01	11	10
00		000	010	100	011	101	110	000	100
01		000	010	100	011	101	110	000	111
11		001	010	100	011	101	110	000	111
10		001	010	100	011	101	110	000	100

$Q_2Q_1Q_0$

Karnaugh map

Output logic for One's- counter controller

			Write address				Read address A				Read address B				ALU operations			Shift operations			
State	Q ₂ Q ₁ Q ₀	IE	WA ₂	WA ₁	WA ₀	WE	RAA ₂	RAA ₁	RAA ₀	REA	RAB ₂	RAB ₁	RAB ₀	REB	S ₂	S ₁	S ₀	Sh ₂	Sh ₁	Sh ₀	OE
s ₀	0 0 0	0	X	X	X	0	X	X	X	0	X	X	X	0	X	X	X	X	X	X	0
s ₁	0 0 1	1	0	0	1	1	X	X	X	0	X	X	X	0	X	X	X	X	X	X	0
s ₂	0 1 0	0	0	1	1	1	0	0	0	1	0	0	0	1	1	0	1	0	0	0	0
s ₃	0 1 1	0	0	1	0	1	0	0	0	1	X	X	X	0	1	1	1	0	0	0	0
s ₄	1 0 0	0	1	0	0	1	0	0	1	1	0	1	0	1	0	0	1	0	0	0	0
s ₅	1 0 1	0	0	1	1	1	0	1	1	1	1	0	0	1	1	0	1	0	0	0	0
s ₆	1 1 0	0	0	0	1	1	0	0	1	1	0	0	0	1	1	0	1	1	1	0	0
s ₇	1 1 1	0	X	X	X	0	0	1	1	1	0	0	0	1	1	0	1	0	0	0	1

$$IE = Q'_2 Q'_1 Q_0$$

$$RAA_2 = 0$$

$$RAB_2 = Q'_1 Q_0$$

$$S_2 = Q_1 + Q_0$$

$$WA_2 = Q'_1 Q'_0$$

$$RAA_1 = Q_2 Q_0$$

$$RAB_1 = Q'_1 Q'_0$$

$$S_1 = Q'_2 Q_0$$

$$WA_1 = Q_2 Q_0 + Q_2 Q'_1$$

$$RAA_0 = Q_2$$

$$RAB_0 = 0$$

$$S_0 = 1$$

$$WA_0 = Q'_1 Q_0 + Q_1 Q'_0$$

$$REA = Q_2 + Q_1$$

$$REB = Q_2 + Q_1 Q'_0$$

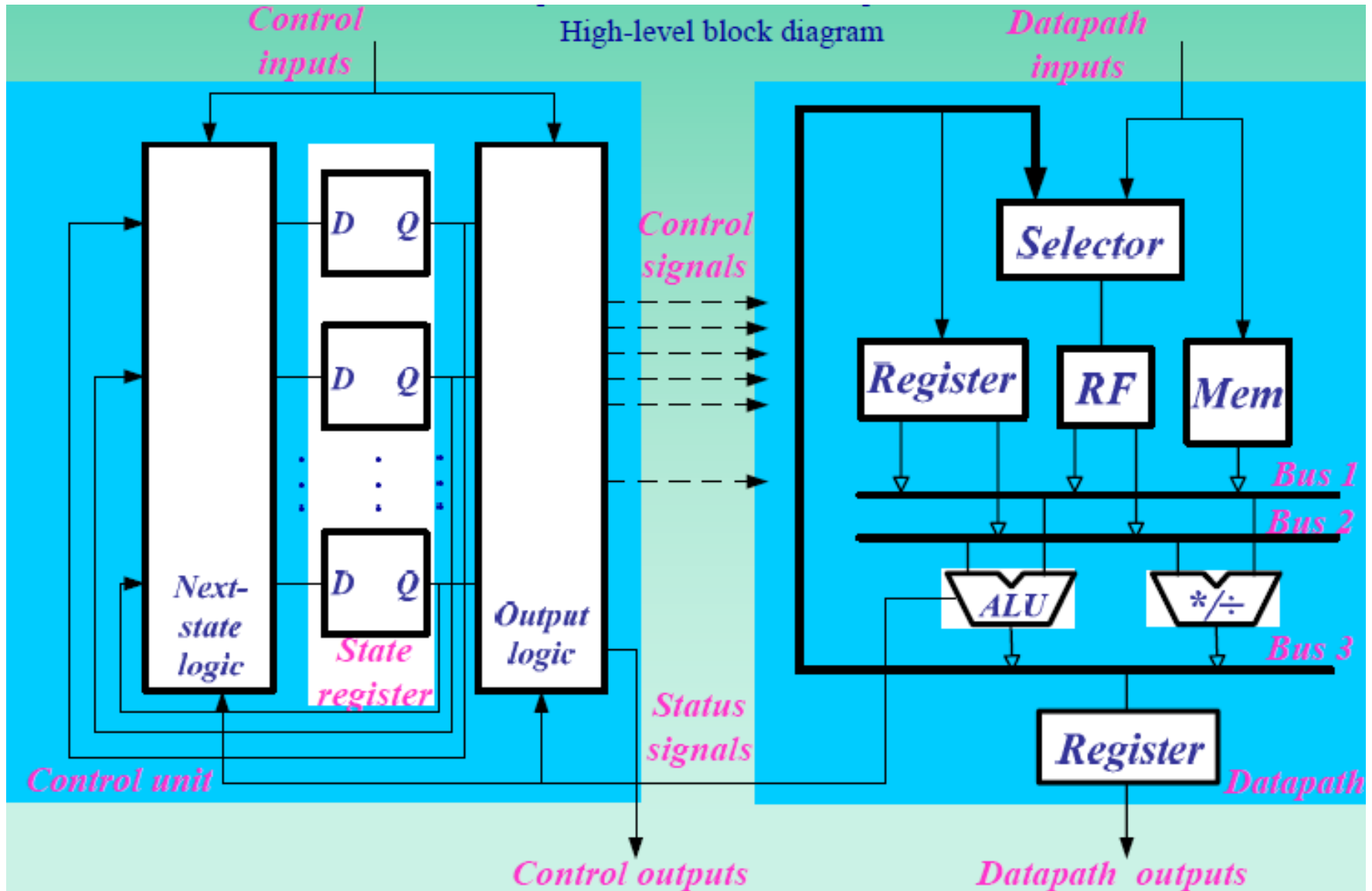
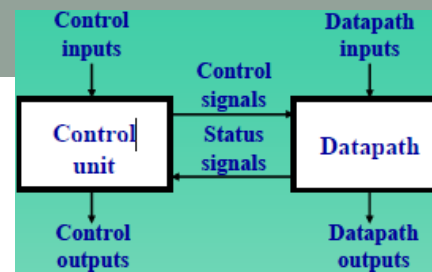
$$Sh_2 = Sh_1 = Q_2 Q_1 Q'_0$$

$$Sh_0 = 0$$

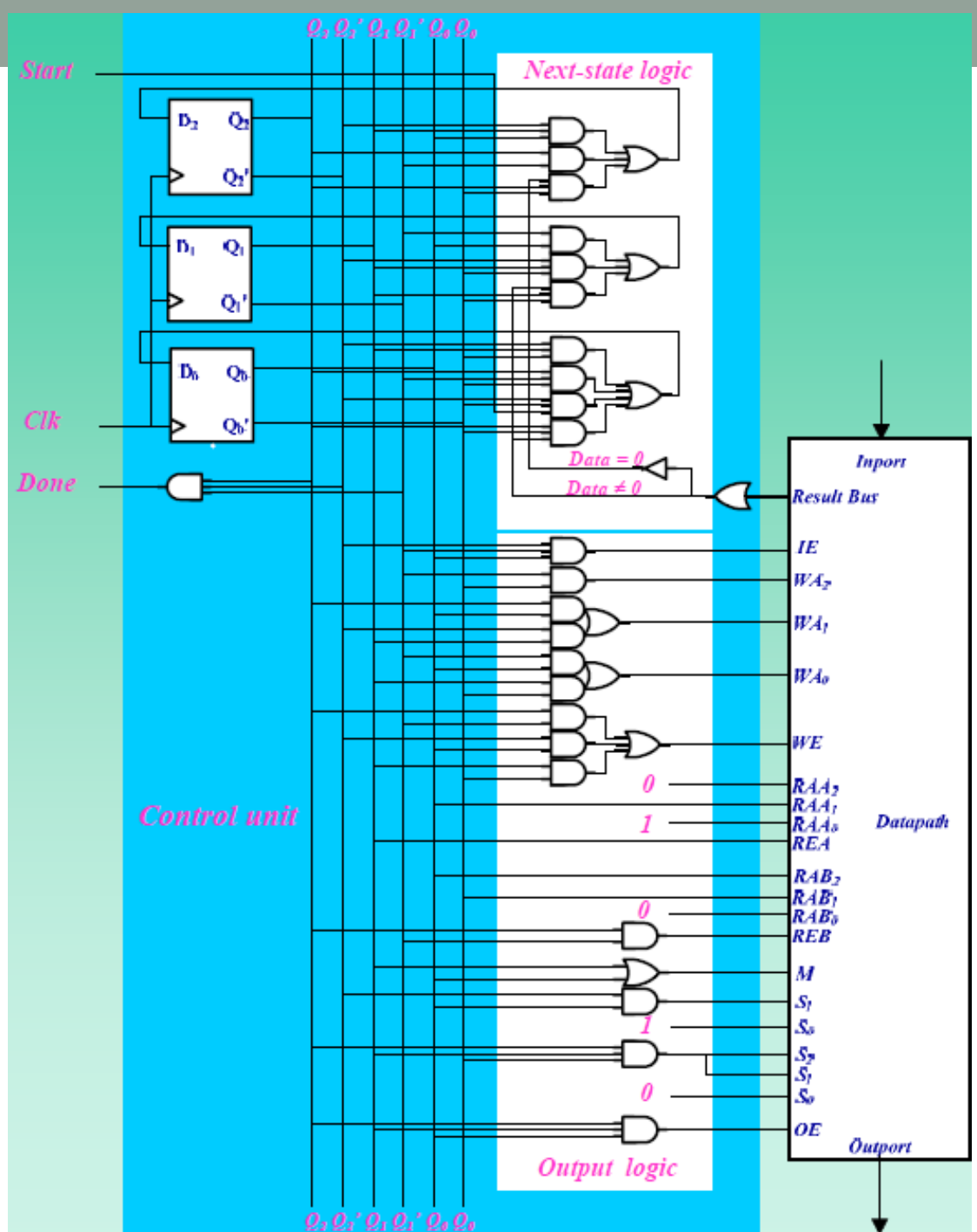
$$WE = Q_2 Q'_1 + Q'_2 Q_0 + Q_1 Q_0$$

$$OE = Q_2 Q_1 Q_0$$

Design model



One's-counter schematic



Chapter Summary

○ We introduced sequential components

- Simple (registers, counters)
- Storage (register-files, memories)
- Complex (stacks, queues, datapaths and control units)

○ Datapath and control units are used to implement standard processors and application specific integrated circuits
(One's count example)